

Val Araújo Barbosa

Software Engineer | Python Backend | API Integrations | AI/RAG

Rio de Janeiro, Brazil

[LinkedIn](#) | [GitHub](#)

Professional Summary

Software Engineer with 5+ years of experience in Python development, backend systems, APIs, automation, and cloud-based solutions. I began my software engineering career by building a Python web application from scratch with Django and MongoDB. I later worked with microservices, REST API integrations, endpoint testing, and cloud functions.

In my most recent role, I built enterprise automation workflows that integrated ITSM platforms, Active Directory, Microsoft 365, SQL Server, cloud environments, and patch management tools through APIs. For a large financial institution, these automations handled high volumes of HR-related service requests, reducing repetitive manual work and the risk of human error. I have also expanded my AI knowledge through hands-on projects with LLMs, RAG architectures, hybrid search, vector databases, and retrieval pipelines.

Professional Experience

Software Developer

ITS Soluções December 2024 - July 2026

- Built enterprise automation solutions with Python and PowerShell that integrated APIs, databases, and corporate platforms.
- Developed API-driven automations for the TIVIT Sensr ITSM platform at a large financial institution. The automations handled high volumes of HR-related service requests by orchestrating actions across Active Directory, Microsoft 365, and SQL Server. Tasks included user provisioning, access changes, group membership management, credential disabling, and employee offboarding. This reduced the manual workload for support teams and the risk of human error.
- Built a patch management workflow that retrieved client virtual machine inventories through Qualys APIs, stored VM data in SQL Server, and created patch deployment pipelines for operating system and application patches. The workflow managed API rate limits during large inventory loads through controlled pauses and workflow resumption.
- Automated virtual machine inventory retrieval and disk snapshot creation, listing, and deletion across AWS, Oracle Cloud, and Google Cloud. These workflows supported safe system updates and recovery procedures.

Software Developer

CWI Software June 2022 - August 2024

- Designed and executed test scenarios to validate critical customer purchase flows in a large enterprise's checkout application.
- Created and executed API test scenarios with Postman and performed end-to-end testing, log analysis, and troubleshooting.
- Worked on application integrations with legacy systems.
- Developed and maintained data integration and ETL workflows with Dremio and Airflow.

Junior Software Engineer

Facily Solutions and Technology November 2021 - April 2022

- Developed and maintained Python microservices with Flask and FastAPI.
- Built and consumed REST API endpoints for communication between backend services.
- Developed Google Cloud Functions and worked with applications running on Google Cloud Platform (GCP).
- Applied Clean Architecture principles to improve code organization, maintainability, and scalability.

Junior Python Developer

Tectia Information Technology and Services September 2020 - February 2022

- Built a Python web application from scratch with Django and MongoDB, taking it from initial business requirements through development and implementation.
- Designed MongoDB data models and developed the backend functionality required by the application.
- Worked directly with stakeholders to understand business needs and translate them into technical solutions.

Innovation Consultant — ALI Program

Sebrae / CNPq June 2015 - November 2017

- Supported approximately 40 micro and small businesses through Sebrae's ALI (Local Innovation Agents) Program, working with them to identify opportunities for innovation, process improvement, and technology adoption.
- Worked directly with business owners to assess operational challenges and guide the adoption of new processes, technologies, and management practices.
- Encouraged participating companies to adopt practical changes aimed at improving efficiency and competitiveness while developing a stronger culture of innovation.

Education

MBA in Strategic Business Management — Anhanguera University, 2014–2015

Bachelor's Degree in Production Engineering — FUMEC University, 2008–2013

Training

Data Science & AI Bootcamp — Le Wagon | 400-hour intensive hands-on program **Automotive Technician Program — SENAI Belo Horizonte** | 2008–2009

Certifications

Goethe-Zertifikat B1 — German | Certified ScrumMaster (expired)

Languages

Portuguese — Native | English — Upper-Intermediate (B2) | German — B1